(i)

Let v_A be the final speed of particle A, and v_B be the final speed of particle B.

Assume the particles are colliding elastically. By conservation of momentum and kinetic energy, we have:

$$bmu = mv_A - bmv_B \tag{1}$$

$$\frac{1}{2}bmu^2 = \frac{1}{2}bmv_B^2 + \frac{1}{2}mv_A^2 \tag{2}$$

Note: The equation (1) implicitly defines v_B as the speed in the opposite direction of the initial velocity u.

From equations (1) and (2), we can simplify:

$$bu = v_A - bv_B \implies b(u + v_B) = v_A \tag{3}$$

$$bu^2 = bv_B^2 + v_A^2 \implies b(u^2 - v_B^2) = v_A^2 \tag{4}$$

From equation (4), we factorize to get $b(u + v_B)(u - v_B) = v_A^2$. Substitute equation (3) into this:

$$v_A(u - v_B) = v_A^2 \implies u - v_B = v_A$$

Substitute $v_A = u - v_B$ back into the simplified momentum equation $bu = v_A - bv_B$:

$$\begin{aligned} bu &= (u - v_B) - bv_B \\ bu &= u - (1 + b)v_B \\ (1 + b)v_B &= u - bu \\ v_B &= \left(\frac{1 - b}{1 + b}\right) u \end{aligned}$$

Note: Since $b > 1$, $v_B < 0$. This implies that particle B actually continues to move in the same forward direction after the collision.

Now, substitute v_B to find v_A:

$$\begin{aligned} v_A &= u - v_B \\ &= u - \left(\frac{1 - b}{1 + b}\right) u \\ &= \left(\frac{1 + b - (1 - b)}{1 + b}\right) u \\ &= \left(\frac{2b}{1 + b}\right) u \end{aligned}$$

Proof that $v_A < 2u$:
Since $b > 1$, it follows that:

$$\begin{aligned} & \frac{b}{1+b} < 1 \\ \implies & 2\left(\frac{b}{1+b}\right) < 2 \\ \implies & \left(\frac{2b}{1+b}\right) u < 2u \quad \blacksquare \end{aligned}$$

(ii)

From part (i), we know that if a moving particle has a mass b times that of the particle it collides with (which is at rest), the velocity of the resting object becomes $\left(\frac{2b}{1+b}\right) u$.

Consider the first collision between B_1 and B_2. The masses are:

$$\begin{aligned} m_{B_1} &= \lambda^{n+1-1} m = \lambda^n m \\ m_{B_2} &= \lambda^{n+1-2} m = \lambda^{n-1} m \end{aligned}$$

Therefore, $m_{B_1} = \lambda m_{B_2}$. Applying the result from part (i) with $b = \lambda$, the velocity of B_2 is:

$$v_{B_2} = \left(\frac{2\lambda}{1+\lambda}\right) u$$

This relationship applies to all subsequent collisions. For B_n, the velocity is:

$$v_{B_n} = \left(\frac{2\lambda}{1+\lambda}\right)^{n-1} u$$

Finally, B_n collides with particle A (mass m). Since $m_{B_n} = \lambda^{n+1-n} m = \lambda m$, we apply the formula one last time:

$$v_A = \left(\frac{2\lambda}{1+\lambda}\right) v_{B_n} = \left(\frac{2\lambda}{1+\lambda}\right)^n u$$

Since $\lambda > 1$, we have $\frac{2\lambda}{1+\lambda} > 1$. As $n \to \infty$, the term $\left(\frac{2\lambda}{1+\lambda}\right)^n \to \infty$, hence $v_A \to \infty$. This proves there is no upper limit on the speed of A.

Case $\lambda = 4$:
We require $v_A > 20u$:

$$\begin{aligned} \left(\frac{2 \cdot 4}{1+4}\right)^n u &> 20u \\ \left(\frac{8}{5}\right)^n &> 20 \end{aligned}$$

Taking $\log_{10}$ on both sides:

$$n \log_{10}\left(\frac{8}{5}\right) > \log_{10} 20$$
$$n > \frac{\log_{10} 2 + 1}{3\log_{10} 2 - (1 - \log_{10} 2)}$$
$$n > \frac{\log_{10} 2 + 1}{4\log_{10} 2 - 1}$$

Using $\log_{10} 2 \approx 0.30103$:

$$n > \frac{1.30103}{1.20412 - 1} \approx 6.37$$

Thus, the least integer value is $\min(n) = 7$.

This solution was submitted by **Roy**

Published and refined by **Roy**

Last updated: April 23, 2026